



VIGHNAX (BhaviṣyAdvaktā)

World-model forecasting of network attack progression

Architecture document · SIH 2026 · PS SIH26153 (NTRO): AI-based Network Attack Forecasting from Network Traffic Data · Team VIGHNAX (Team ID 127364) · open source (MIT), fully offline

54 / 201

attack onsets forecast before they begin (baseline: 17)

0.70 vs 0.67

macro forecast AUROC on unseen test data, vs the baseline

7 of 7

PS datasets in one model, tested on unseen traffic

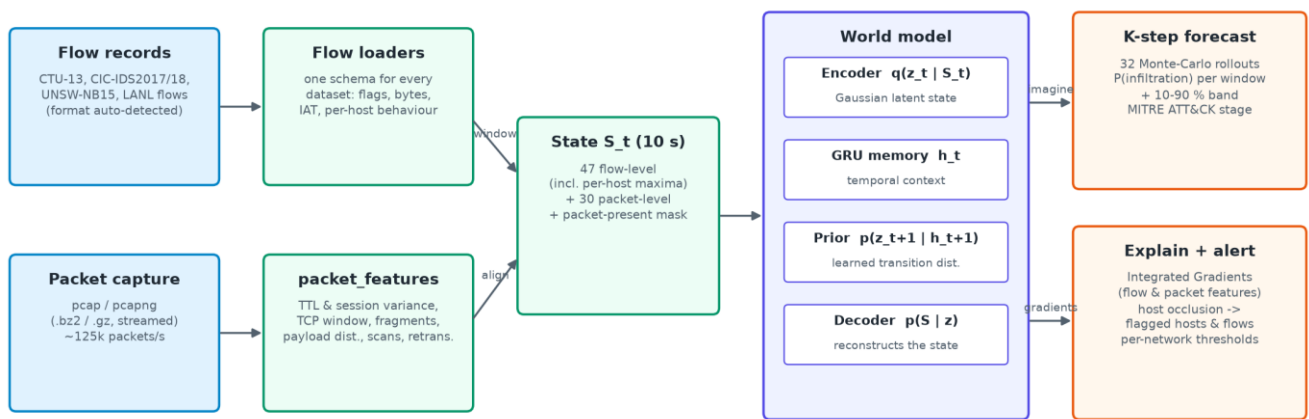
~8 ms

per 10 s window on a laptop CPU, fully offline

1. Idea

Instead of classifying flows one at a time, VIGHNAX learns **how the state of a network evolves** (the transition distribution $P(S_{t+1} | S_t)$) from flow-level and packet-level telemetry. Rolling that model forward 60 s shows whether the current trajectory leads to compromise, which MITRE ATT&CK stage comes next, which traffic features drive the prediction and which hosts are responsible. One model is trained on all seven datasets named in the PS.

2. System architecture



Offline dashboard: live timeline, forecast band, stage mix, attributions, flagged hosts & flows, incident report | no cloud APIs

3. Components

Ingestion	Reads every PS dataset's native format, detected by content: CTU-13 Argus binetflow, CICFlowMeter CSV (CIC-IDS2017/2018), UNSW-NB15 records, LANL flows. Packet captures (pcap / pcapng, .gz / .bz2) are streamed at ~125k packets/s with headers decoded directly; a Scapy reference parser is included.
Network state S_t	Every 10 s window becomes a 78-dim vector: 47 flow-level features (5-tuple statistics, flags, bytes, packets, duration, IAT, fwd/bwd ratios, per-host maxima so one compromised host is not diluted), 30 packet-level features (TTL mean and per-session variance, TCP window, fragment / DF flags, payload-size distribution, SYN-only / RST / ICMP shares, retransmissions, port-scan signatures) and a packet-present mask.
World model	Encoder $q(z_t S_t)$ to a Gaussian latent; GRU memory h_t ; learned transition prior $p(z_{t+1} h_{t+1})$; decoder $p(S z)$. Heads on (h, z) give $P(\text{infiltration})$ and a 7-way ATT&CK stage distribution. Loss = reconstruction + KL(posterior prior) + attack and stage losses on observed states and on imagined K-step rollouts against the true future labels. Dataset-balanced sampling, modality dropout (CSV-only input) and volume augmentation.
Forecast and explanation	32 Monte-Carlo latent trajectories rolled $K = 6$ windows (60 s) ahead: $P(\text{infiltration})$ per window with a 10–90 % band and the predicted stage mix with technique IDs. Integrated Gradients attributes each forecast to flow and packet features against a benign reference state; removing each host in turn and re-forecasting ranks the responsible hosts and flows.
Interface	Offline Flask dashboard (bundled Chart.js and fonts): one drop zone for flow files, PCAPs or a .zip; timeline, forecast fan, stage mix, attributions and flagged hosts update live; per-network alert thresholds (site profiles); one-click incident report; SHA-256-verified model weights.

4. Data and evaluation protocol

- All seven PS datasets:** CTU-13 (flows + full packet captures), CIC-IDS2017 and CSE-CIC-IDS2018 (CICFlowMeter flows), UNSW-NB15 (flow records), LANL cyber1 (router flows + red-team events), DARPA 1999 (tcpdump + attack truth list) and CICIOT2023 (per-activity packet captures).
- Test data unseen in training:** other botnet families (CTU-13), other days (CIC, UNSW, DARPA), another 12 h slice (LANL), other attack types (CICIOT2023). Validation = every 5th 5-minute block of the training sources.

- **Fair baseline:** logistic regression on the same state features and targets. Operating thresholds are chosen per dataset on validation only (most detection with at most 5 % false alarms), with the same rule for both models. Labels map to ATT&CK stages by behaviour; all seven stages occur.

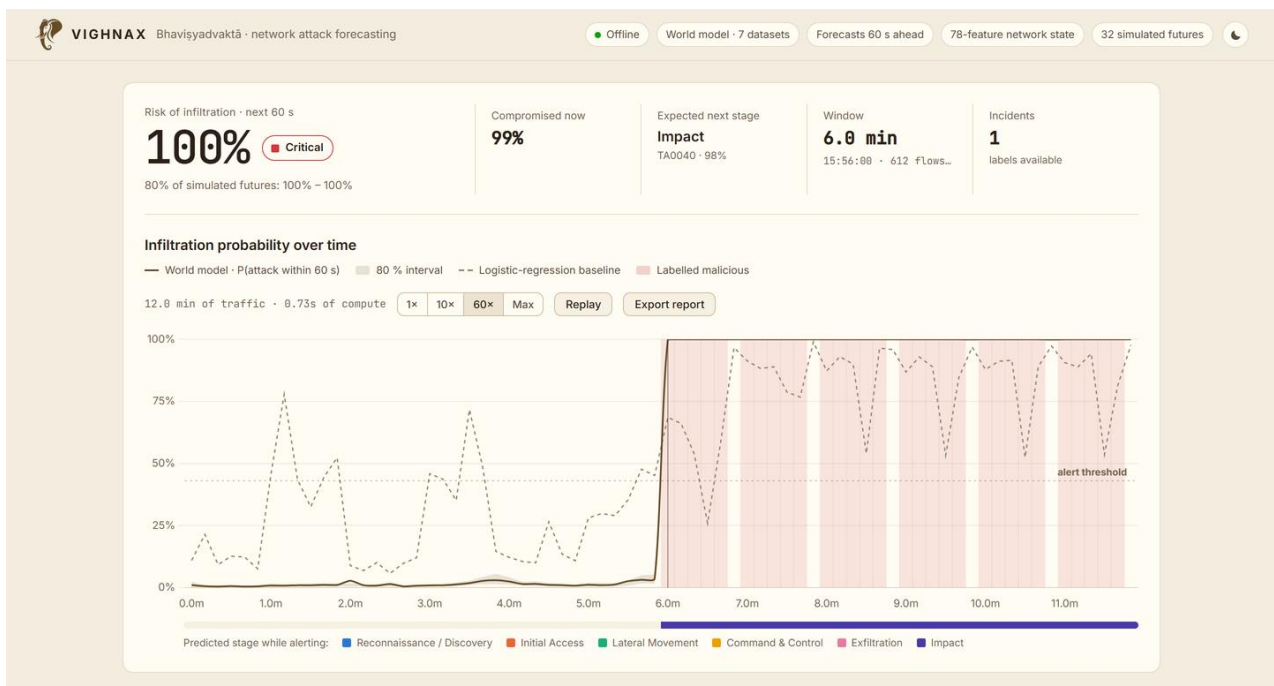
5. Results on unseen test data

Dataset (unseen test data)	Forecast	Early warning	Detection
CTU-13	0.83 / 0.60	0.82 / 0.48	0.89 / 0.74
CIC-IDS2017	0.56 / 0.56	0.47 / 0.47	0.65 / 0.65
CSE-CIC-IDS2018	0.83 / 0.60	0.57 / 0.55	0.83 / 0.62
UNSW-NB15	0.95 / 0.99	0.63 / 0.71	1.00 / 0.95
LANL cyber1	0.37 / 0.56	0.34 / 0.57	0.40 / 0.55
DARPA 1999	0.63 / 0.69	0.60 / 0.61	0.69 / 0.81
CICIoT2023	n/a	n/a	0.99 / 0.76
Macro average (six temporal datasets)	0.70 / 0.67	0.57 / 0.57	0.74 / 0.72

AUROC, world model / logistic-regression baseline. Early warning = forecasting from windows that are still benign. CICIoT2023 episodes are composed (benign capture, then attack capture), so only detection is scored.

- **Early warning:** 54 of 201 attack onsets were forecast before they began, against 17 for the baseline; when the model warns, it does so 51–60 s ahead on average.
- **Forecasting:** ahead on CTU-13 and CSE-CIC-IDS2018, tied on CIC-IDS2017; the baseline is ahead on UNSW-NB15, LANL cyber1, DARPA 1999. On validation the world model leads on every dataset, so generalising to unseen days is the main open problem.
- **Detection:** ahead on 4 of 7 datasets; the baseline is better on LANL cyber1 and DARPA 1999.
- **Cost:** ~8 ms per 10 s window streamed on a laptop CPU (32 rollouts), 0.04 ms batched on a GPU; ~125k packets/s ingestion.

6. Working prototype



Offline dashboard on an unseen CIC-IDS2017 day: risk stays low for six minutes, then goes critical as the DDoS starts; the baseline (dashed) raises false spikes.

7. Deployment, limits and next steps

- **Deployment:** air-gapped and CPU-only; streams captures in bounded memory. Reproducible from one seeded configuration: `fetch_datasets.sh` → `prepare_data.py` / `prepare_all.py` → `train.py` → `evaluate.py`.
- **Limits:** weak on LANL-type authentication attacks and behind the baseline on DARPA 1999; alert thresholds need per-site calibration; PCAP-only input relies on flows rebuilt from packets.
- **Next:** host-graph (GNN) state, online threshold calibration, a streaming NetFlow / IPFIX collector and SOC integration (syslog, STIX).

Code, weights and benchmark: <https://github.com/NAMAN-THAKUR-1944/SIH2026-SIH26153-VIGHNAX> · **Demo video:** https://youtu.be/pPTe-d_TByU